B.M.S. College of Engineering

Bio-inspired Algorithms | 23DS7BSBIO

Concrete Compressive Strength Prediction Using Differential Evolution–Optimized Support Vector Regression

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Introduction to Optimization Problems

Objective and search space

Optimization selects parameters that minimize an objective.

Here: minimize five-fold development RMSE of an RBF-SVR.

Decision variables: $\log_{10}(C)$, $\log_{10}(\gamma)$, $\log_{10}(\varepsilon)$.

Constraints: fixed parameter bounds and 300 objective calls.

The objective is evaluated by fitting models; derivatives are unavailable.

Holdout performance evaluates fixed selections after tuning.

What are Nature-Inspired Algorithms?

Nature-inspired search

Search methods motivated by biological, collective or physical processes.

They explore candidate solutions and use measured fitness to guide updates.

Genetic Algorithm: selection, crossover and mutation.

Differential Evolution: mutation from population-vector differences.

Particle Swarm Optimization: individual and social search information.

No gradient is required; performance still depends on task and controls.

Classification of Nature-Inspired Algorithms

Algorithm families

Evolutionary methods: Genetic Algorithms and Differential Evolution.

Swarm methods: Particle Swarm Optimization and Ant Colony Optimization.

Physical-process methods: Simulated Annealing.

This project implements DE/rand/1/bin for continuous parameters.

Random Search is the equally budgeted tuning comparator.

The other algorithms are background methods, not measured baselines.

Problem Statement

Prediction task

Predict concrete compressive strength in MPa from ingredients and age.

SVR accuracy depends on regularization C, kernel γ and tolerance ε.

Question: does DE improve tuning under an equal evaluation budget?

Compare default SVR, Random Search–SVR and DE–SVR.

Keep data splits, preprocessing, search bounds and budgets fixed.

Report all predefined seeds and the limits of the comparison.

Objectives and Scope

Research questions

RQ1. Does tuning improve a default RBF-SVR? (frozen study)

RQ2. Does DE beat equally budgeted Random Search? (frozen study)

RQ3. How much does the input representation matter? (ln(age))

RQ4. Does DE keep an edge in a 10-D search space and on new splits?

Scope: one UCI dataset, one RBF-SVR, explicit DE/rand/1/bin.

Extensions were pre-registered before they were run.

Dataset and Preprocessing

Dataset and evaluation split

UCI Concrete Compressive Strength: 1,030 rows and eight inputs.

Seven ingredient quantities (kg/m³) and curing age (days).

Target: compressive strength (MPa); no missing values.

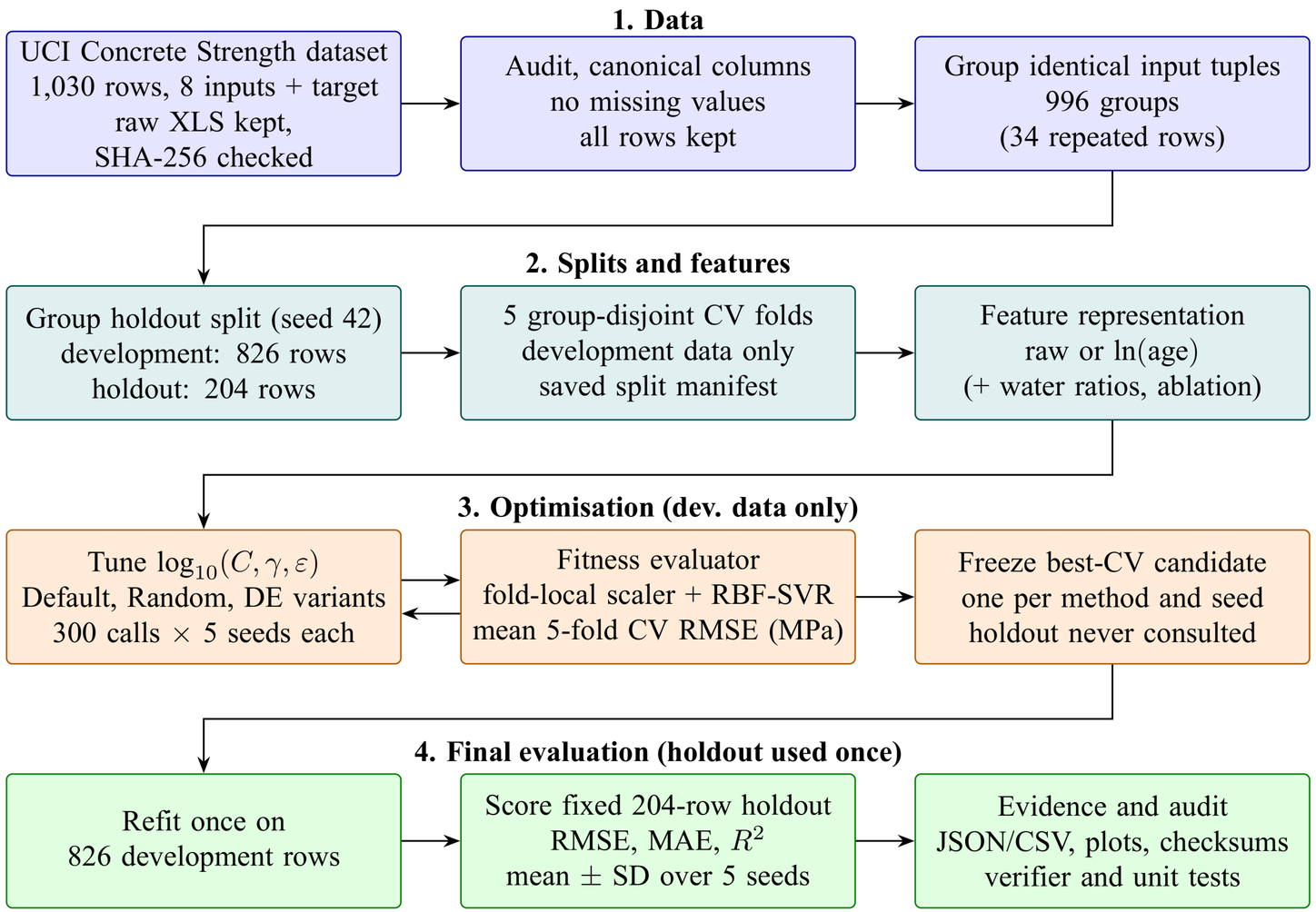
996 unique input tuples; identical inputs stay in the same group.

Development: 826 rows / 796 groups. Holdout: 204 rows / 200 groups.

Five group-disjoint CV folds; StandardScaler fitted within each fold.

Frozen runs use original features; the extension uses ln(age). All rows kept.

System Architecture: Data to Evidence



Four stages: data, splits and features, development-only optimisation, single holdout evaluation.

Implementation and Results

Pipeline

Raw-data audit and saved group splits

StandardScaler → RBF-SVR

Development CV evaluates candidates

Refit selected parameters on development

Evaluate on fixed holdout

Save histories, predictions and provenance

DE configuration

Population: 15; F = 0.8; CR = 0.9

300 calls including initialization

19 trial generations after initialization

Seeds: 11, 22, 33, 44, 55

$\log_{10}$ C: [−2, 4]; $\log_{10}$ γ: [−5, 1]

$\log_{10}$ ε: [−3, 1]

Differential Evolution: Explicit Operators

DE/rand/1/bin

1. Initialize 15 candidates uniformly within the log bounds.
2. Select three distinct donors, all excluding the current target.
3. Mutate: $v = x_{r1} + 0.8(x_{r2} - x_{r3})$; clip to bounds.
4. Binomial crossover at 0.9; force one mutant coordinate.
5. Evaluate trial using mean development CV RMSE.
6. Accept trial when no worse; defer updates until generation end.

Stop after exactly 300 objective calls. No hidden polishing.

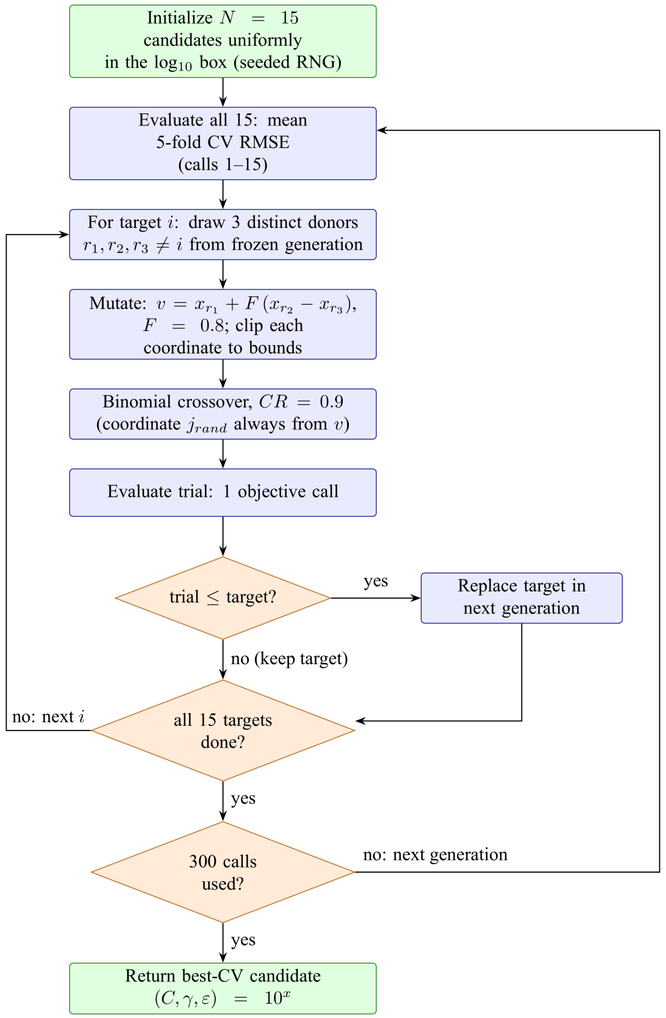
DE Flowchart (as Implemented)

Per generation

15 trial candidates

Donors from the frozen generation

Greedy acceptance: trial ≤ target



Budget

15 initial calls

19 generations × 15 trials

= 300 calls per seed

Frozen-generation donors, clipping, forced crossover coordinate, greedy selection, 300-call budget.

Performance Metrics and Benchmarking

Same 204 holdout rows; tuned values are mean ± sample SD over five seeds.

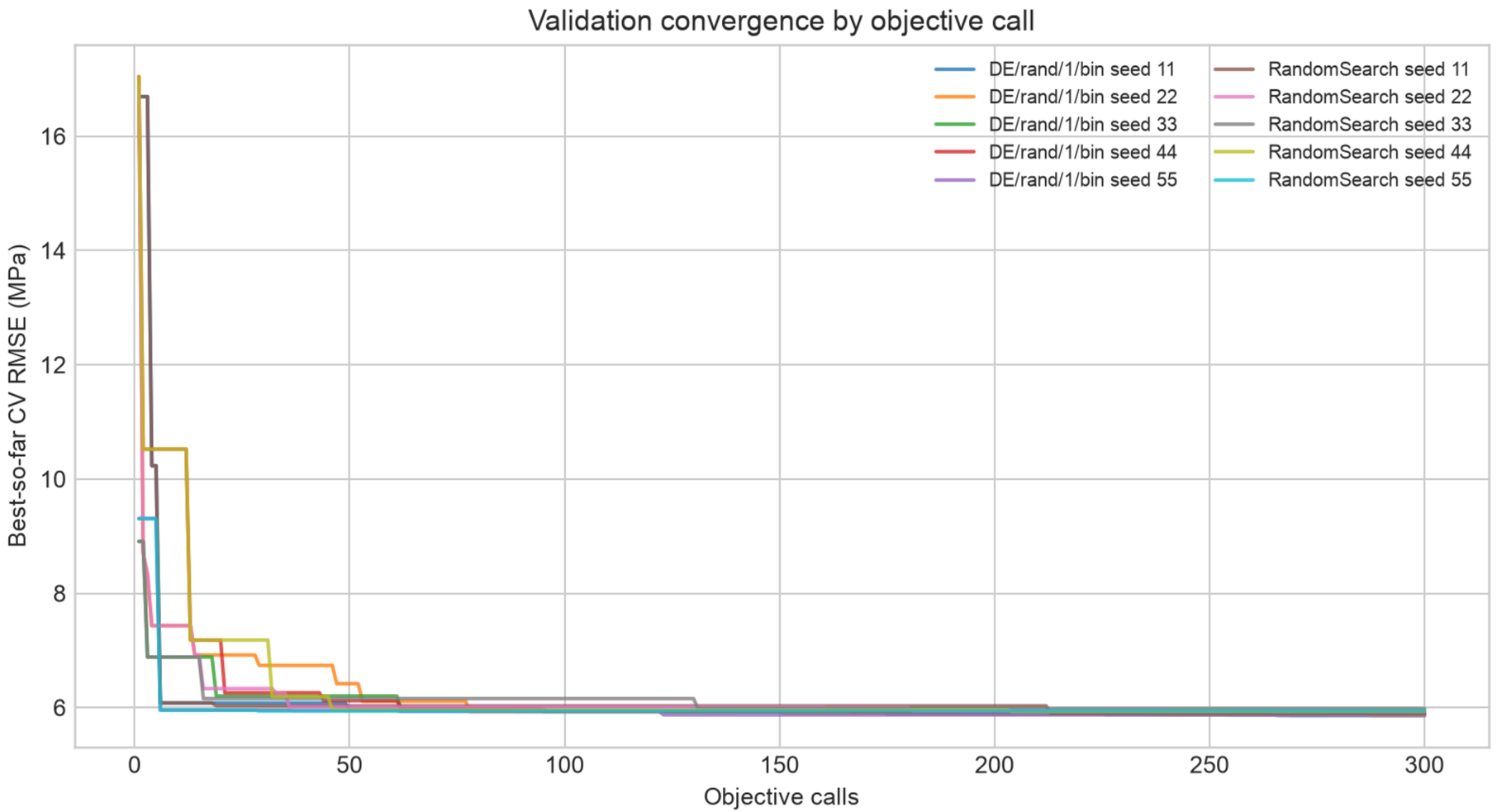
Model	RMSE (MPa)	MAE (MPa)	R^2
Default SVR	9.9069	7.8854	0.5755
Random Search–SVR	6.2328 ± 0.1194	4.3502 ± 0.0253	0.8319 ± 0.0064
DE–SVR	**6.2250 ± 0.1104**	**4.3143 ± 0.0526**	**0.8324 ± 0.0060**

DE reduces RMSE by about 37.2% relative to default SVR.

DE and Random Search differ by only 0.0078 MPa (about 0.1%).

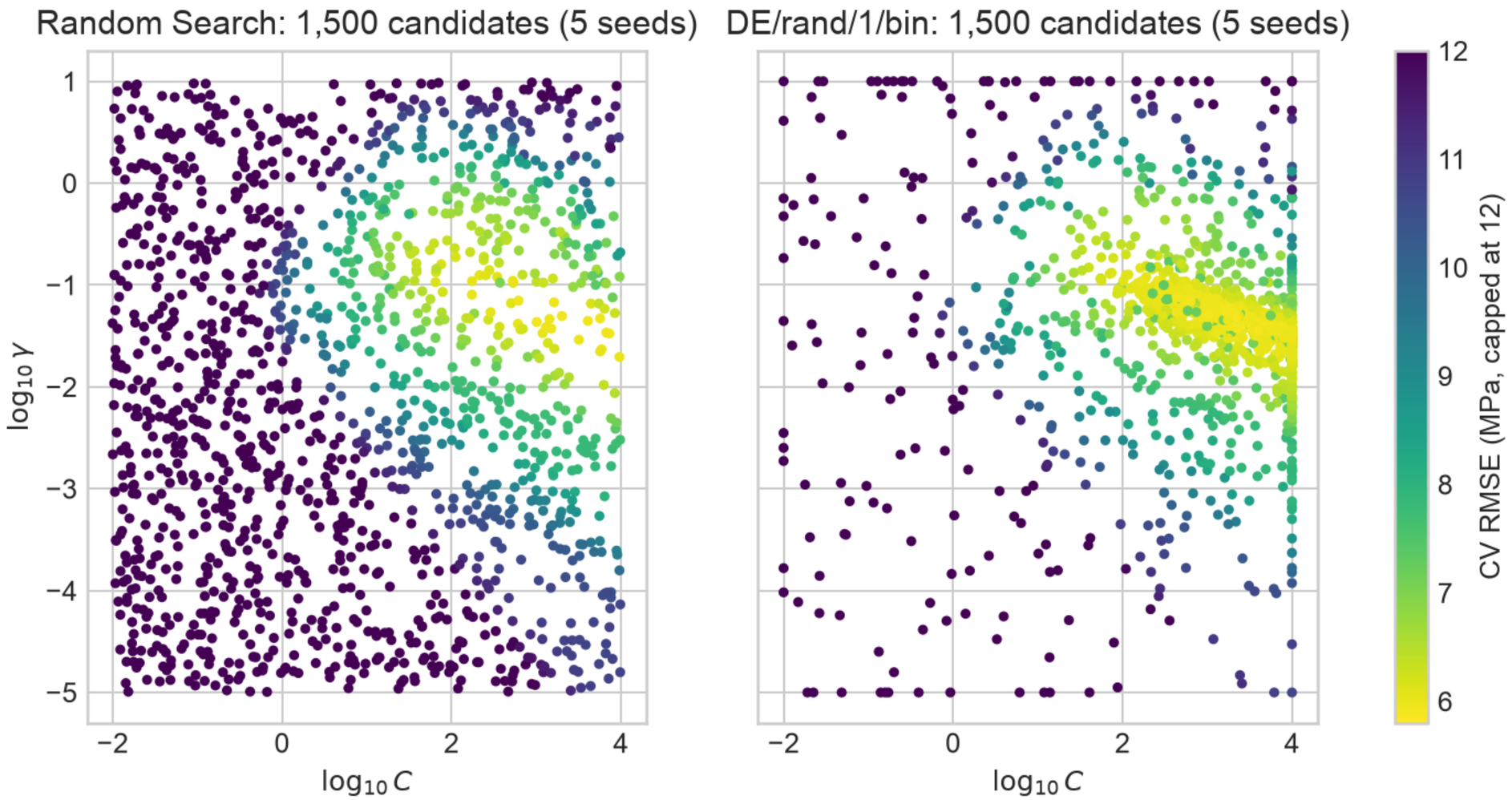
Conclusion: tuning helps; the two searches are essentially equivalent here.

Development Validation Convergence



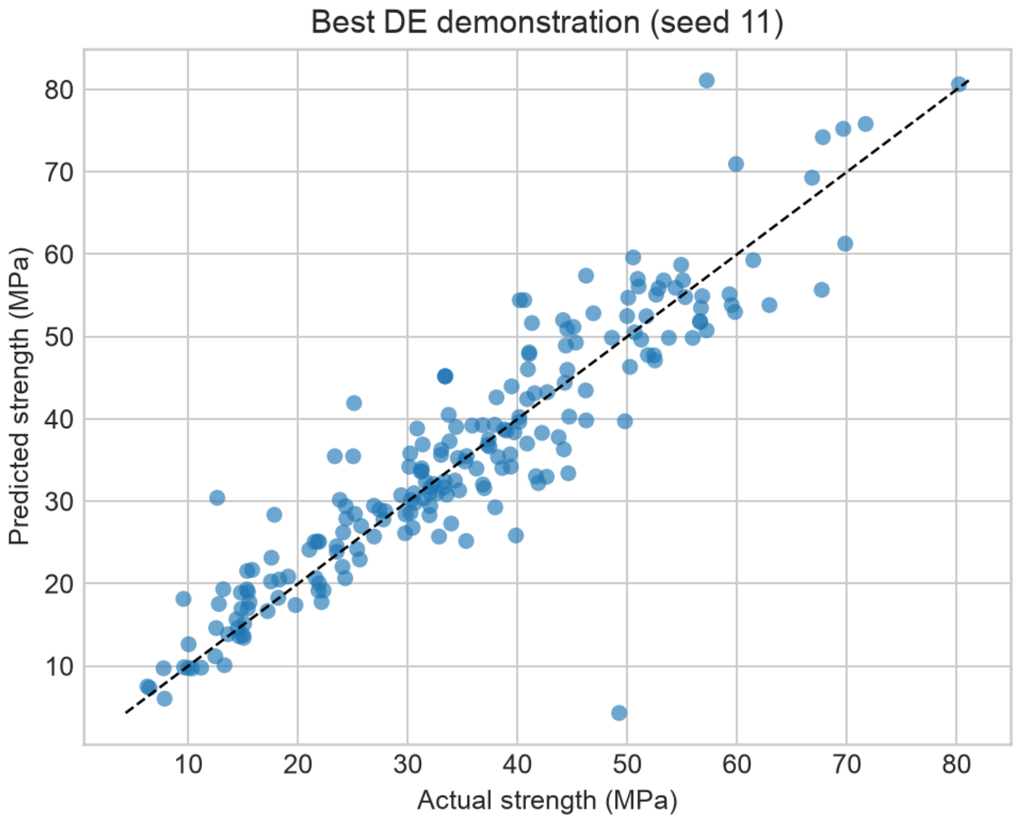
Equal budgets: 300 calls per method per seed, including initialization.

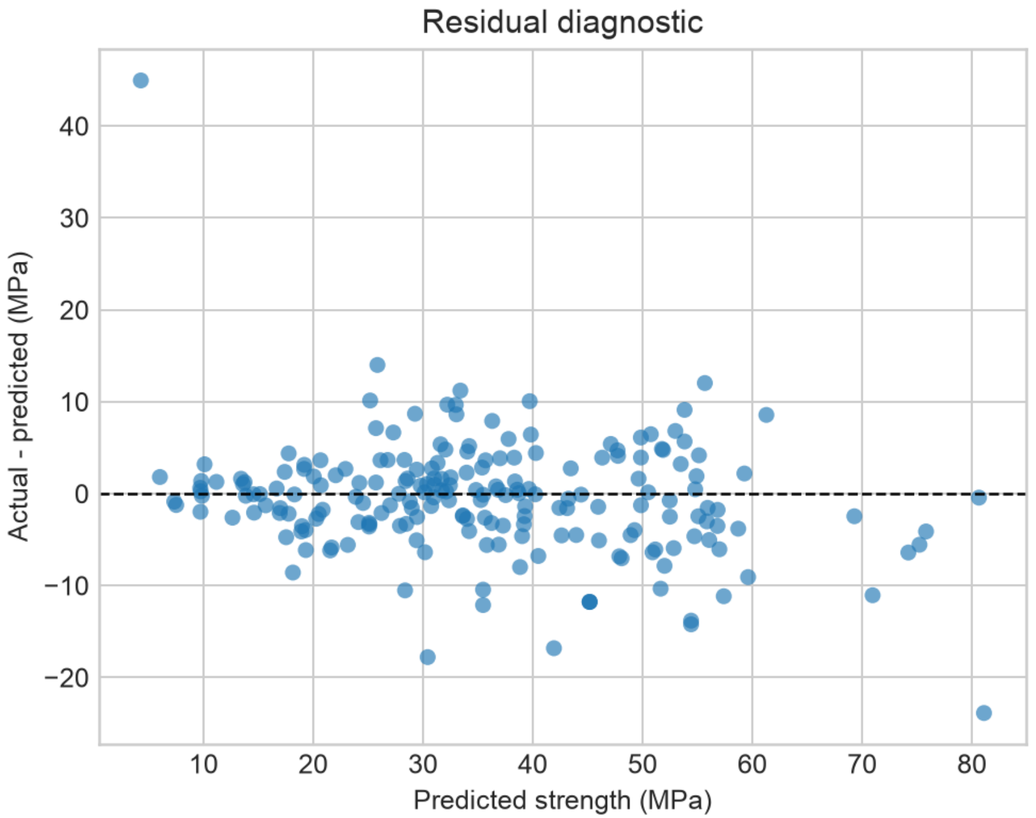
Where Each Search Evaluated Candidates



1,500 candidates per method (5 seeds): DE concentrates in the good region; Random Search stays uniform.

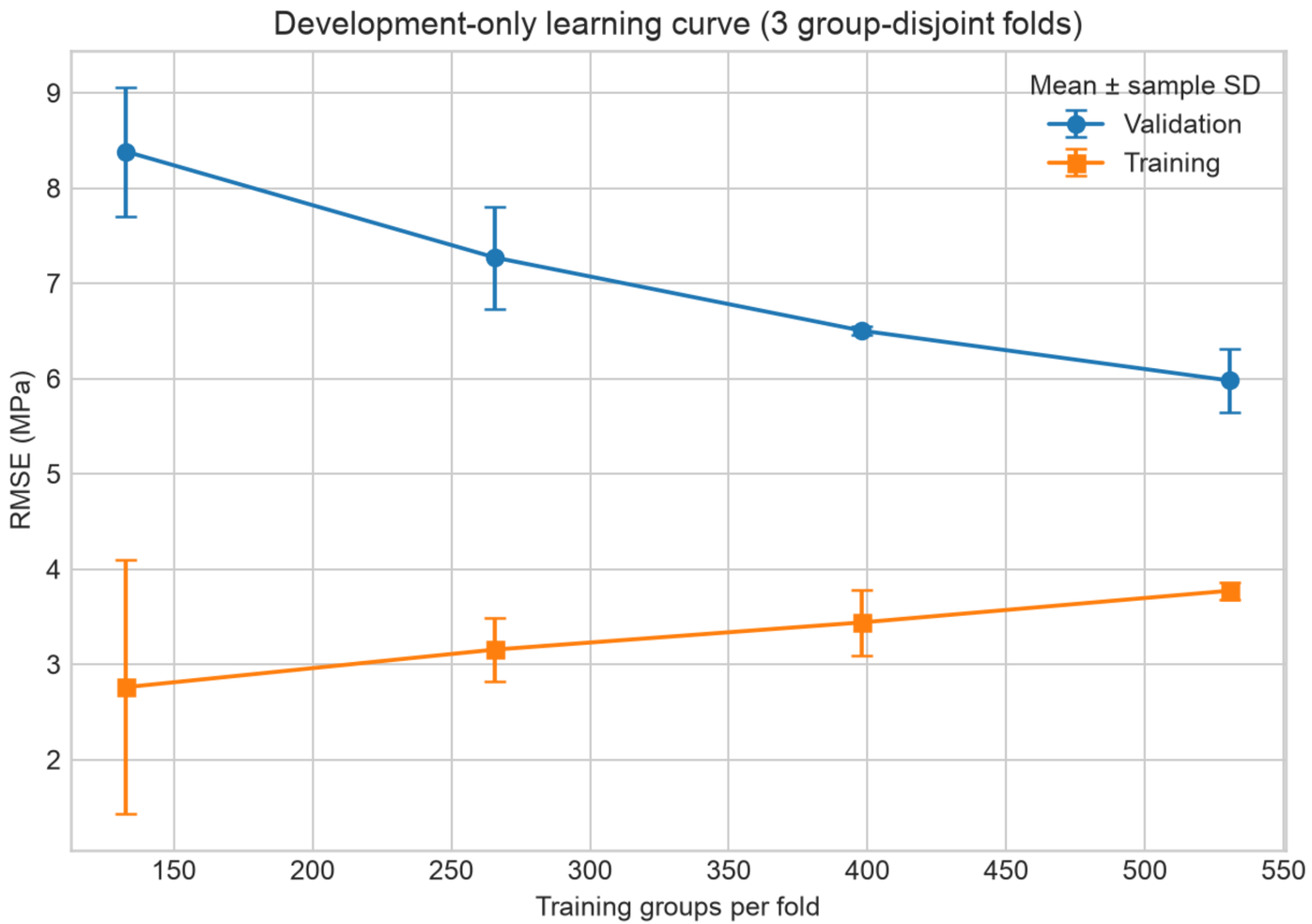
Predictions and Residuals





Demonstration: DE seed 11, chosen by validation; holdout RMSE 6.3712 MPa.

Learning-Curve Diagnostic



Three development-only folds; nested whole groups; fixed parameters; no retuning.

Extension: ln(age) Input for All Three Methods

Same 204 holdout rows; tuned values are mean ± sample SD over five seeds.

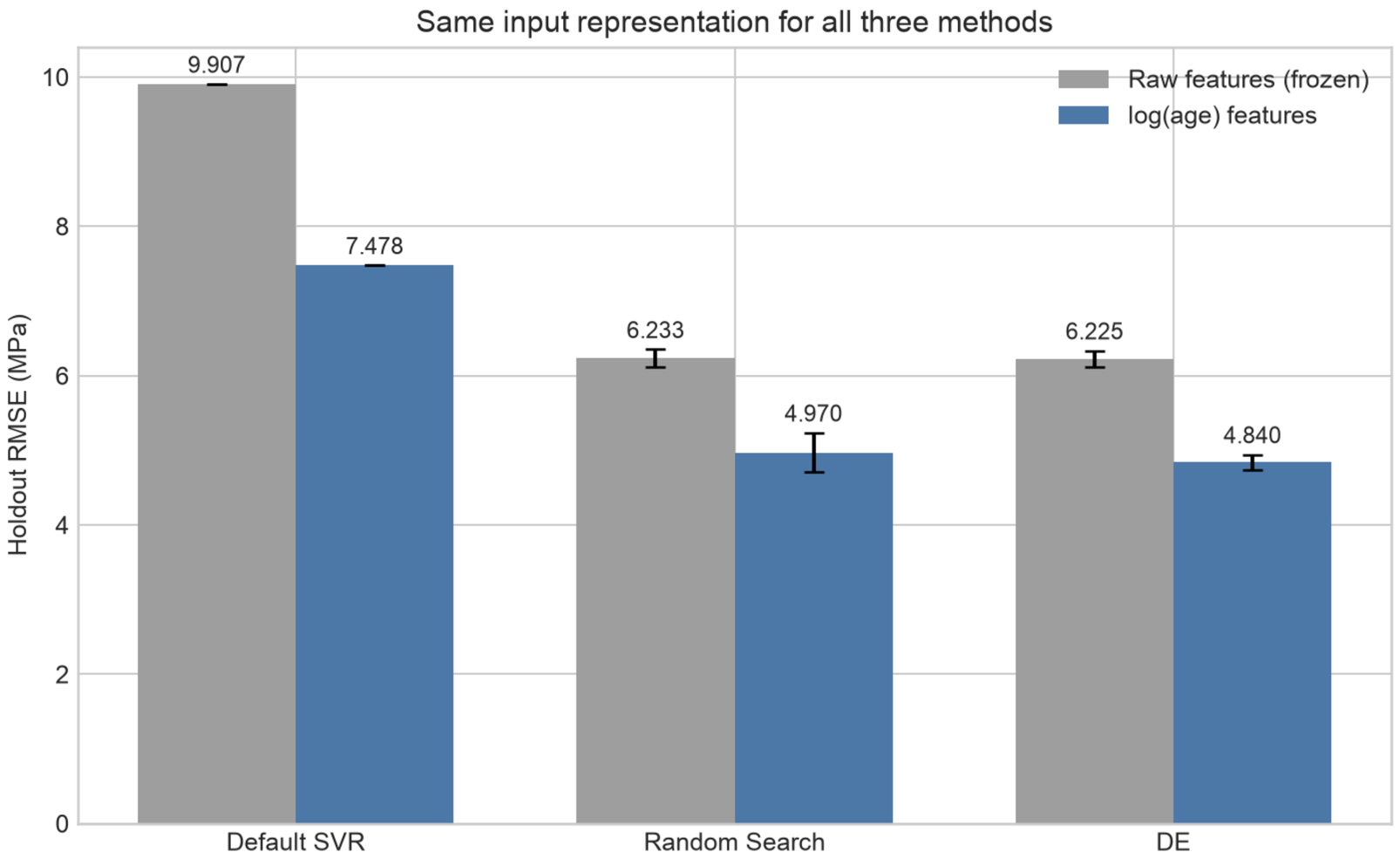
Model	Raw RMSE (MPa)	ln(age) RMSE (MPa)	ln(age) R^2
Default SVR	9.9069	7.4780	0.7581
Random Search–SVR	6.2328	4.9700 ± 0.2656	0.8929
DE–SVR	6.2250	4.8402 ± 0.1039	0.8987

Representation drives the gain: DE 6.225 → 4.840 MPa (22.2% lower).

DE vs Random Search on ln(age): 0.130 MPa apart; no significance claim.

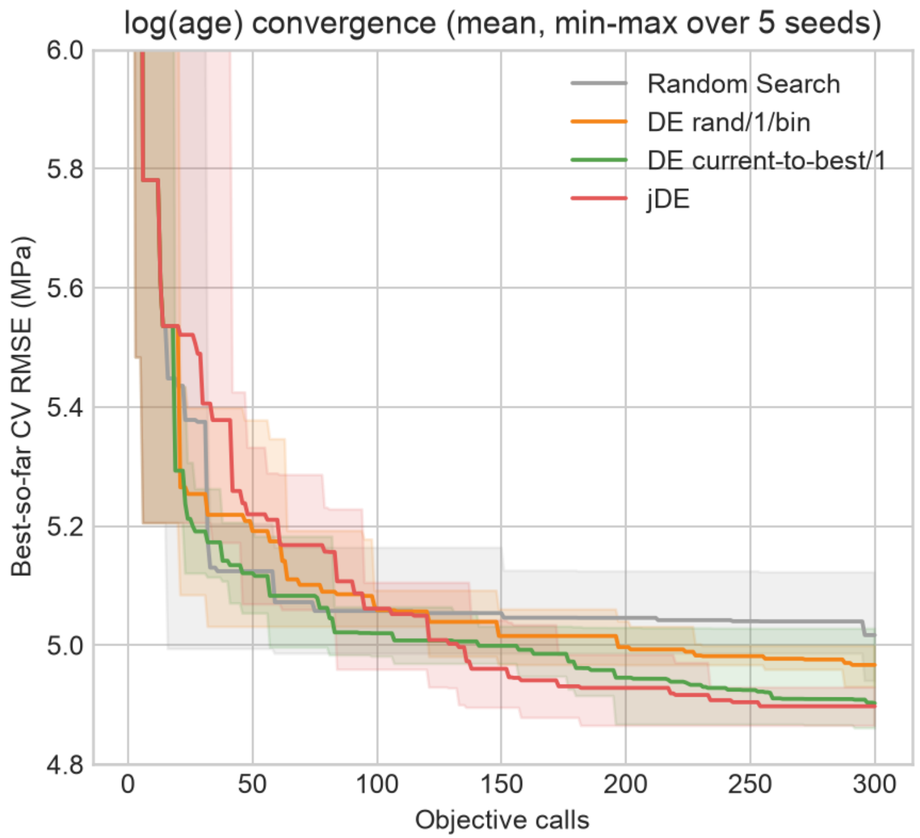
ln(age) chosen after a development-only probe; holdout was inspected in Stage 1.

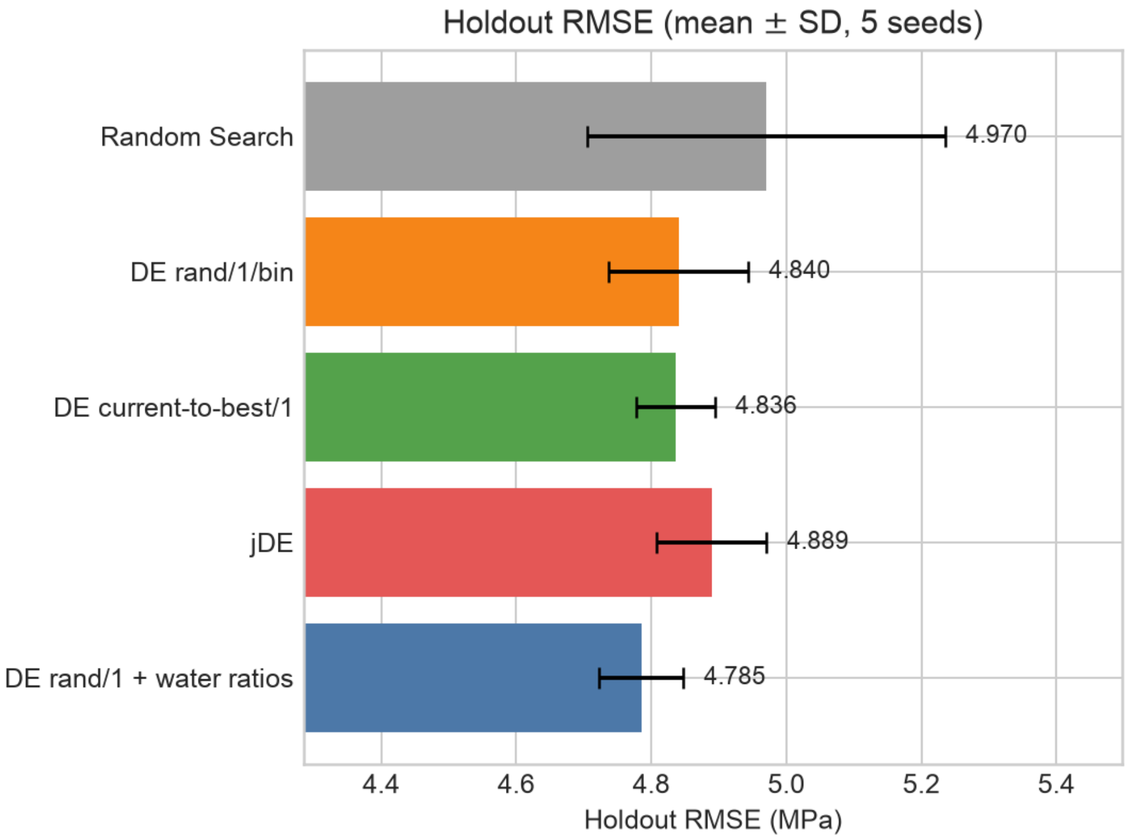
Representation Comparison



Holdout RMSE, mean ± SD over five optimizer seeds; raw rows are the frozen results.

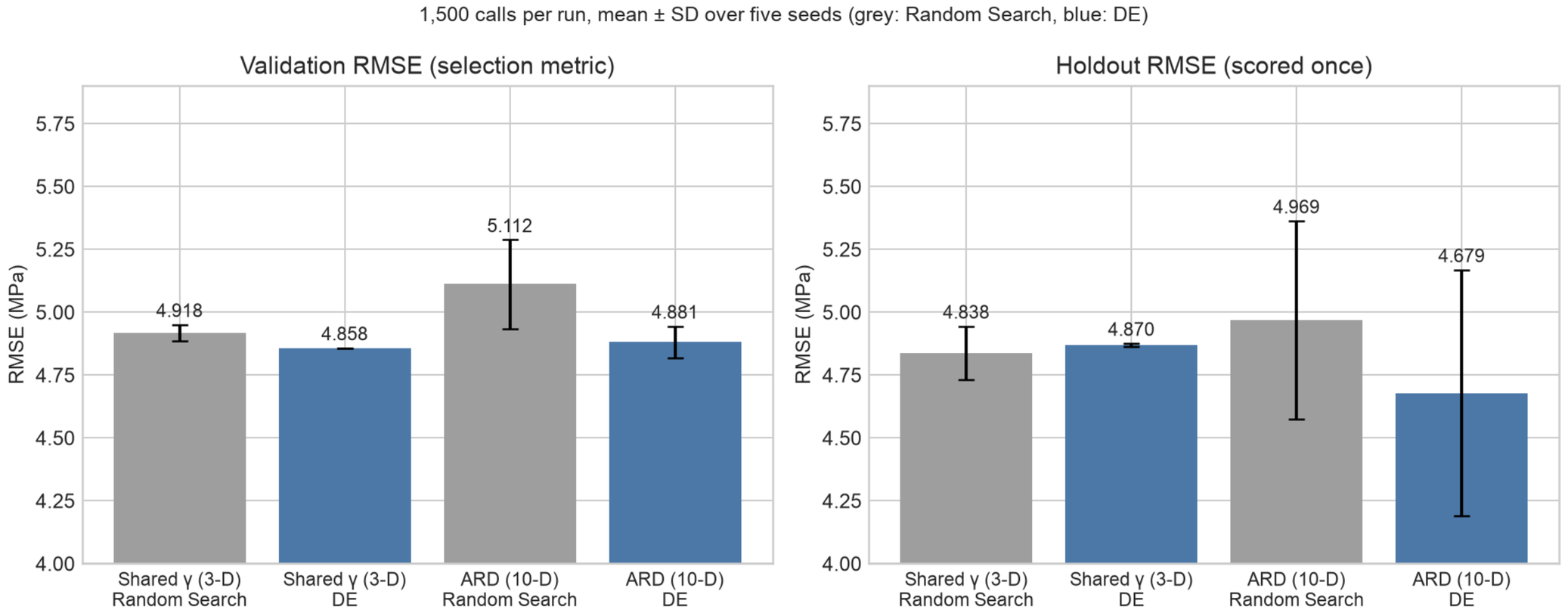
DE Variant Ablation (ln(age))





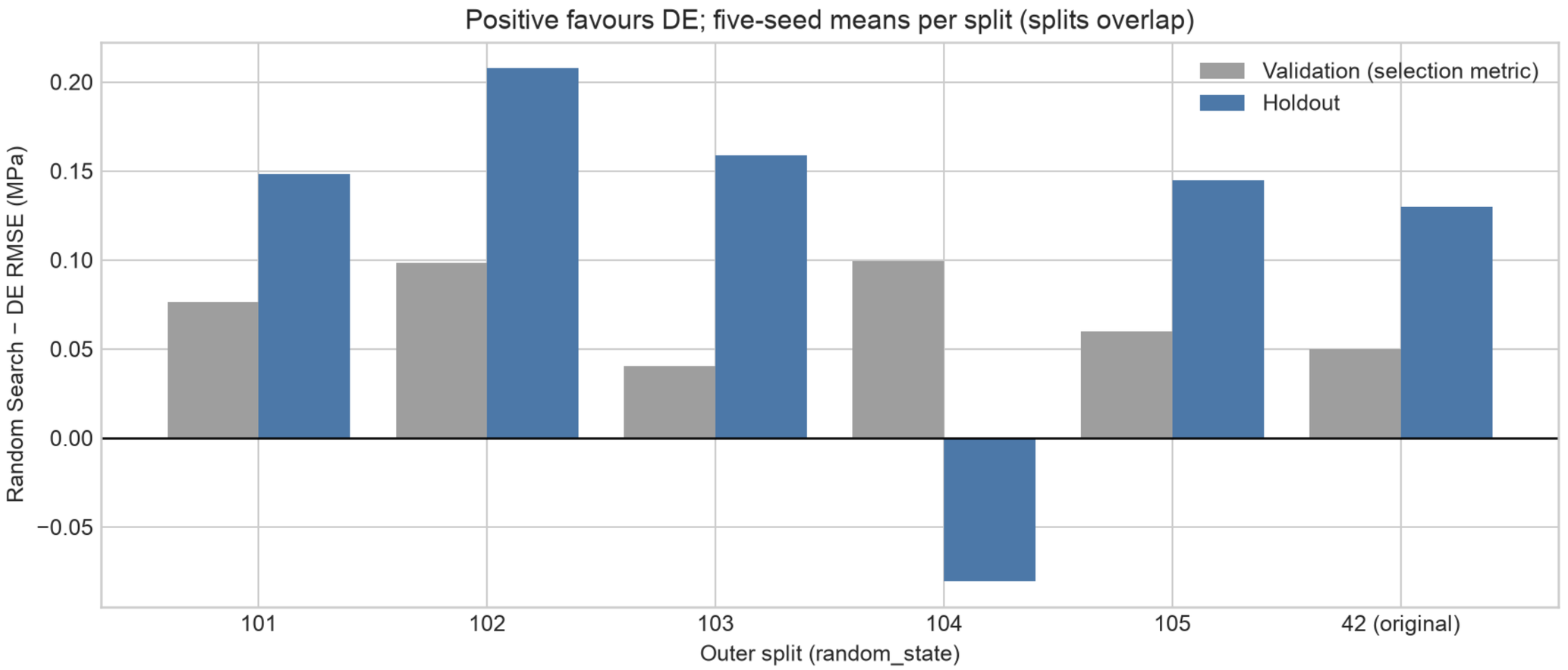
Same 300-call budget; rand/1/bin vs current-to-best/1/bin vs jDE vs Random Search.

Extension 2a: ARD Kernel (10-D Search Space)



1,500 calls, five seeds; y-axis starts at 4.0 MPa. ARD does not improve validation RMSE over a shared γ.

Extension 2b: Replication on Five New Splits



DE lower validation RMSE on 6/6 splits, holdout on 5/6; mean holdout gap 0.118 MPa (indicative; splits overlap).

Key Findings

Summary of evidence

Tuning works: DE cuts holdout RMSE 37.2% (9.907 → 6.225 MPa).

On frozen features DE and Random Search are 0.0078 MPa apart.

Representation matters most: ln(age) lowers DE to 4.840 MPa (22.2%).

DE keeps a small edge of about 0.1 MPa on 5 of 6 splits; no significance claim.

ARD (10-D) gave no accuracy gain; Random Search degrades there, DE does not.

Advantages

What this approach offers

Explicit, tested DE operators for a three-parameter continuous search.

Both tuned methods substantially improve default SVR.

Group-aware splits and fold-local scaling reduce leakage risks.

Equal candidate budgets make the search comparison auditable.

Complete histories, predictions and locked dependencies support reproduction.

Fifty-seven passing tests, checksummed evidence and a saved audit support reproduction.

Limitations

What the evidence cannot establish

One public dataset; one main holdout (Extension 2 adds five overlapping splits).

Seed SD describes optimizer variability, not independent dataset uncertainty.

DE does not show meaningful superiority over equally budgeted Random Search.

Baseline and Random Search holdout evaluations preceded final DE evaluation.

No feature-importance or significance study; ln(age) and ARD came from development probes.

Finite search cannot prove global optimality; no field deployment is validated.

Future Scope

Next experiments

Evaluate on external laboratory data using a frozen model.

Use non-overlapping outer splits or nested group cross-validation.

Test DE control settings systematically under matched budgets.

Study uncertainty and age-stratified errors with a declared protocol.

Add interpretation studies without using holdout feedback for tuning.

Assess material or environmental benefits only with suitable new evidence.

References

Selected sources

[1] I.-C. Yeh, Concrete Compressive Strength, UCI, 1998.
Dataset DOI: 10.24432/C5PK67.

[2] I.-C. Yeh, Cement and Concrete Research, 28(12), 1797–1808, 1998.

[3] R. Storn and K. Price, Journal of Global Optimization, 11, 341–359, 1997.

[4] J. Bergstra and Y. Bengio, JMLR, 13, 281–305, 2012.

[5] A. J. Smola and B. Schölkopf, Statistics and Computing, 14, 199–222, 2004.

[6] scikit-learn documentation: SVR, StandardScaler and GroupKFold.

THANK YOU

Tuning helps, and the input representation helps most.

DE’s edge over Random Search is small (about 0.1 MPa) and untested for significance.

Questions and discussion

github.com/R-V-Abhishek/concrete-strength-de-svr